

Matthew Ison

Dr. Chen

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Hospital Distance: Effect on Heart Disease Prevalence

Introduction:

Access to healthcare is one of the pillars of health economics and plays a vital role in determining health outcomes. One important aspect of healthcare access is the transactional costs associated with receiving treatment. This can take the form of the costs of transportation to healthcare facilities, such as hospitals, which impact the ability of a patient to receive adequate care. This is particularly true for individuals with chronic conditions such as heart disease. Heart disease is a leading cause of death and mortality worldwide, making it a significant public health concern. Thus, understanding if distance to a hospital plays a role in heart disease prevalence is essential for our understanding on how to improve healthcare access and outcomes for patients.

This paper examines the effect of hospital distance on heart disease prevalence. Specifically, we want to investigate whether census tracts located closer to hospitals have lower heart disease prevalence compared to census tracts farther away. This analysis hopes to add to existing literature on healthcare access, by providing empirical evidence on the relationship between hospital distance and heart disease prevalence within the state of Idaho. Secondly, this paper has implications for policy, as understanding this relationship can direct resources to improve healthcare outcomes for communities who need it.

In order to understand the effect of distance on heart disease, this paper uses a both geographic information system (GIS) and Ordinary least squares (OLS) as its quantitative methods for analysis. The hypothesis is that individuals living closer to hospitals are more likely to receive adequate care for the causes of heart disease, due to the reduction of transactional cost of transportation, thus leading to a reduction of prevalence of heart disease. Essentially, do individuals forego treatment at a hospital because of the costs of access thus increasing the severity of the heart issue thus contributing to the prevalence of heart disease? By testing this hypothesis, we aim to provide estimates on how distance to hospitals plays a role in chronic illnesses, specifically heart disease, and how we can reduce the burden of these illnesses on our communities.

Literature Review:

Healthcare Access and Health Outcomes

Previous studies have highlighted the importance of healthcare access in determining health outcomes. In their systematic review of studies covering the effect distance to healthcare has on health outcomes for adults in the global north, Kelly et al, 2016, found that 77% of the 133 studies analyzed had found evidence of a relationship between the two. Similarly, Graves, 2008, conducted a review on literature pertaining to geographic information system's role in examining healthcare access and health outcomes. The review highlighted two particular papers that emphasized the importance of distance to healthcare facilities. Luther et al's, 2003 paper measured the effect access to primary care facilities had on health outcomes, particularly among racial minorities. The authors categorized zip codes as high or low access based on access to clinics within the area. Ultimately, the authors found that racial minorities from high-access

areas had lower rates of chronic illnesses than did minorities from low-access areas. The second paper, by Parker and Campbell 1998, surveyed patients in Scotland about their access to healthcare services. Firstly, it was found that the use of services decayed as the distance increased grew between the place of residence of the patients and the location of services. Additionally, patients were seen to have a lack of perception of the actual distance to healthcare services, believing they were closer than in reality. Which gives insights into the perception of ease of access to care among individuals, which can influence utilization of healthcare resources.

Role of access on heart disease

Studies have also done research specifically on the role access to healthcare facilities has on heart disease only. Khan et al, 2022, looks at the barriers to care for patients diagnosed with congenital heart disease. In the paper they find that, distance to care centers was associated with a decreased probability of follow-ups, however emergency room and outpatient visits overall were associated positively with increased distance from these centers. This may indicate that patients wait until the disease is in critical condition before trying to access care. Another paper by Yamashita and Kunkel, 2010, focused on heart disease mortality and the impact of hospital access in Ohio counties. Using bivariate regression, the authors find that when controlling for socio-economic factors, the mortality rate for heart diseases and distance to hospitals relationship is not statistically significant. They argue that distance is not as much as a contributing factor to mortality rate. Rather, the potential utilization of the hospitals around the patients are a more important factor. A critical idea to consider for our own analysis as we look at prevalence of heart disease in Idaho and its relationship to hospital distance.

Role of Healthcare Infrastructure on economic development

In the field of health economics, there is an emphasis to understand the impacts that community health and health infrastructure have on our general economy. Studies have shown evidence supporting the idea that increases in healthcare infrastructure are associated with healthier populations and higher levels of economic growth (Varkey et al, 2020). Additionally, Kawai et al, 2017, and Nazmul et al, 2014, show the negative impacts of health and chronic illness on economic productivity. In a more granular analysis, Ona and Davis, 2011, conduct a quasi-experimental analysis on a critical access hospital program on Kentucky communities. They find the program, designed to provide healthcare access to rural communities, benefited communities economically compared to communities that did not participate in the program.

Data:

The data for this paper was collected from a variety of United States government dataset releases and merged together using Idaho census tract FIPS codes.

Socioeconomic variables were collected from the 2019 American Community Survey (ACS) 5-year estimates through the Census API. The ACS provides detailed information on demographic, social, economic, and housing characteristics for the U.S. population. The 5-year estimates are used here (compared to 1 or 3 years available) to provide reliable data for smaller geographic areas, as this paper uses a smaller geographic resolution i.e. census tracts. The use of shorter year estimates is often unreliable at that small area of study.

PM2.5 (particulate matter with a diameter of 2.5 micrometers or less) census tract daily concentrations for 2019 were collected from the Centers for Disease Control and Prevention

(CDC). PM2.5 concentrations are a measure of air pollution and can have significant impacts on health, including heart disease. These daily concentrations were averaged for the year and attributed to each census tract.

The centroid coordinates for Idaho census tracts were collected from the Census Centers of Population dataset. The centroid coordinates represent the population center of each census tract and are used as location data for the census tracts for our spatial analysis and to calculate distance from closest hospital.

The Idaho hospital coordinates were collected from the Homeland Infrastructure Foundation-Level Data (HIFLD) database through the geospatial management office of the U.S. Department of Homeland Security. These coordinates provide the locations of hospitals in Idaho and are used to calculate the distance of hospitals to census tracts centroids. These coordinates do not contain coordinates for nursing homes or health centers.

The census tract health data, including health outcomes (like population adjusted prevalence of heart disease), health prevention data, health risk behaviors data, and health status data, comes from the CDC PLACES: Local Data for Better Health, Census Tract Data 2023 release dataset. This dataset provides information on various health factors at the census tract level, allowing for our analysis of health outcomes in Idaho census tracts. It should be noted that this dataset uses pre-2020 Census, census tracts FIPS codes. Idaho gained more census tracts in the 2020 census, therefore variables that require the use of FIPS codes merging all are from the most recent pre-2020 data. Additionally, one census tract was not included in the dataset from the CDC and is therefore not included in this analysis.

Additionally, a shapefile used to represent the 2019 location and outlines of census tracts for the state of Idaho was collected from the Census Bureau.

Below is a list of the variables in the dataset and their descriptions:

Description of variables

Name	Variable Label
population	POPULATION
latitudecentroid	LATITUDE-Centroid
longitudecentroid	LONGITUDE-Centroid
FIPSID	Census Tract ID
distancemin	Distance From Closest Hospital
pm25	Average PM2.5 Concentrations in 2016
med_inc	Median Income
med_age	Median Age
pop_female	percentage of population female
pop	Population
avg_house_size	Average Size of Household
educ_total	Total Education Attainment
educ_hs	Education Attainment High School Level
educ_somecollege	Education Attainment Some College Level
educ_asso	Education Attainment Associate Degree Level
educ_bach	Education Attainment Bachelors Degree Level
educ_mast	Education Attainment Master's Degree Level
educ_prof	Education Attainment Professional Degree Level
rent_income_rat	Rent to Income Percentage
rentincome_35_39	Rent to Income higher than 30-39.9
rentincome_40_49	Rent to Income higher than 40-49.9
rentincome_50	Rent to Income higher 50
lackaccess2hi	Prevalence of Lack of Health Insurance
binge	Prevalence of Binge Drinking
bphigh	Prevalence of High Blood Pressure
bpmed	Prevalence of Blood Pressure Medication
heartdisease	Prevalence of Heart Disease
checkup	Prevalence of Routine Checkups
cholscreen	Prevalence of Cholesterol Screening
csmoking	Prevalence of Smoking
depression	Prevalence of Depression
diabetes	Prevalence of Diabetes
fair_poor_health	Prevalence of Fair or Poor Health
highchol	Prevalence of High Cholesterol
leasurephyact	Prevalence of no leisure-time physical activity
obesity	Prevalence of Obesity
phyhealthnotgood14	Prevalence of physical health not good more than 14 days
sleep7hrs	Prevalence of sleeping less than 7 hours

Below is a descriptive statistics table for the variables within the dataset:

Descriptive Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
FIPSID	297	1.603e+10	27716053	1.600e+10	1.609e+10
distancemin	297	5.851	7.097	.169	46.476
pm25	297	7.149	1.246	4.369	9.9
med inc	297	55769.741	16524.586	23179	125387
med age	297	38.453	7.671	21	60.3
pop female	297	2884.559	1926.472	140	14048
pop	297	5783.582	3829.633	264	28516
educ total	297	3755.993	2395.176	218	18449
educ hs	297	829.549	513.162	7	3395
educ somecollege	297	662.162	431.716	44	2730
educ asso	297	369.02	276.103	13	1996
educ bach	297	697.572	657.749	26	5274
educ mast	297	234.963	238.919	0	1783
educ prof	297	56.973	70.251	0	459
rent income rat	297	49.283	53.381	0	313
rentincome 35 39	297	42.219	51.842	0	420
rentincome 40 49	297	53.468	66.157	0	517
rentincome 50	297	123.37	153.56	0	1645
lackaccess2hi	297	10.454	3.838	3.8	28.5
binge	297	15.369	1.694	11.7	22.7
bphigh	297	30.811	4.973	12	42.6
bpmed	297	71.14	5.876	28.4	79.5
heartdisease	297	5.971	1.444	1.1	10.4
checkup	297	71.897	2.645	60.6	78.3
cholscreen	297	79.076	5.497	46.4	89
csmoking	297	16.208	3.282	7.8	30.4
depression	297	23.692	2.311	18.4	36.3
diabetes	297	10.02	2.344	2.7	17.2
fair poor health	297	15.769	3.739	8.3	29.9
highchol	297	32.515	3.919	14.7	40.3
leasurephyact	297	23.175	4.872	12.6	36.1
obesity	297	33.512	3.988	25.2	45.8
phyhealthnotgood14	297	12.1	2.145	7.2	19.2
sleep7hrs	297	30.444	2.134	24.2	38.7
avg house size	297	2.661	.402	1.5	5.15

Methods:

GIS Analysis for Calculating Minimum Hospital Distance

GIS modules in Python were used to help calculate the distances between the centroid coordinates of census tracts and all the hospitals in Idaho. These modules account for certain

things such as the curvature of the earth and coordinate reference systems that pertain to certain geometries, among other things, to form an accurate calculation of distance. From there, I found the minimum distance in miles that it took for each census tract centroid point to get to a hospital point. For this analysis, I used straight-line distance (Euclidean distance) rather than distance taken from a car or through use of road-networks. Optimally, analysis on this subject would factor in those transportation routes, as it could shorten or lengthen the distance to a hospital. However, for this analysis we will simplify that factor in order to obtain a general understanding of the relationship between our two variables.

OLS Multiple Regression Analysis

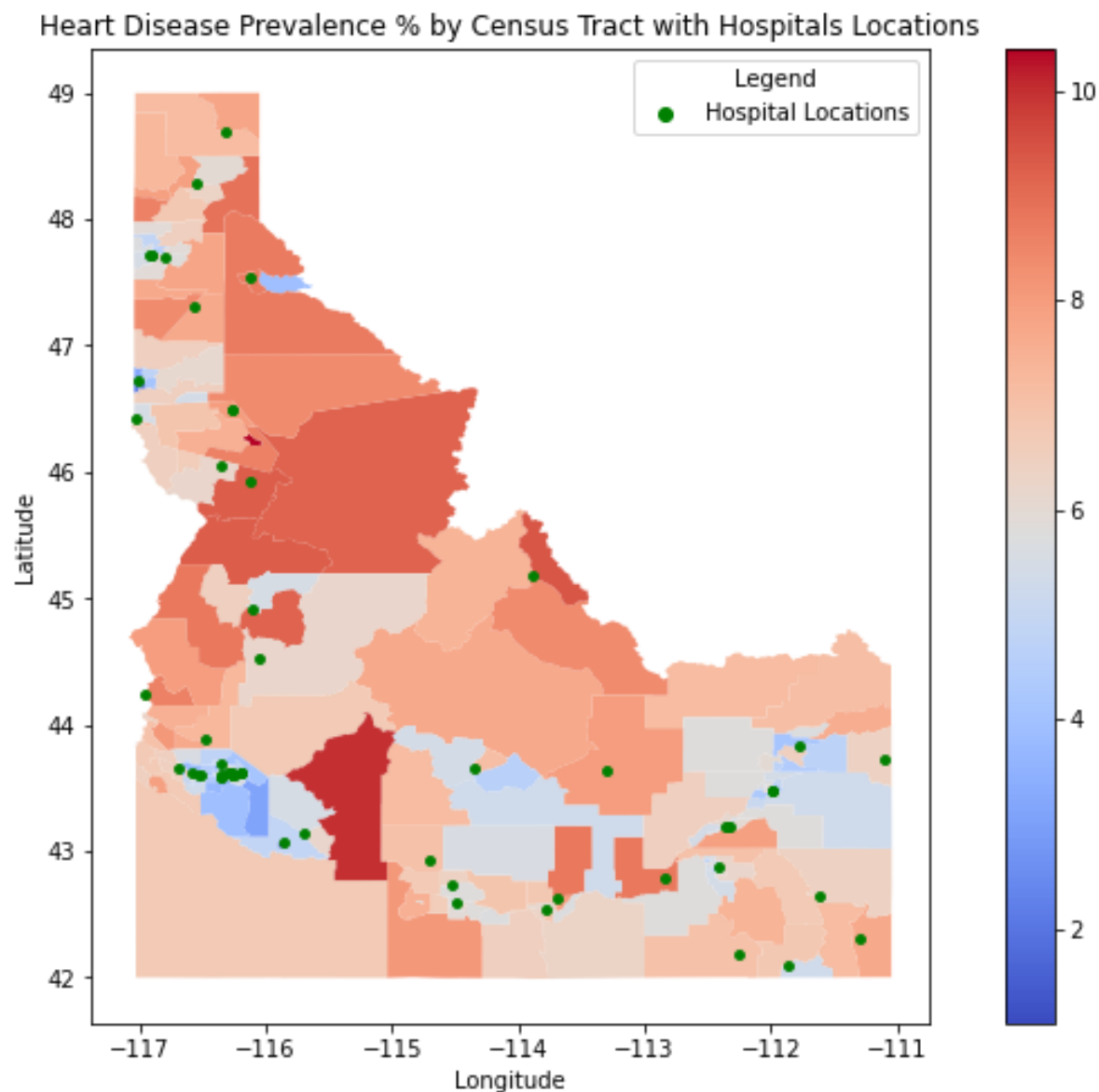
OLS (Ordinary Least Squares) multiple regression analysis was conducted to examine the effect of minimum hospital distance on heart disease prevalence across multiple model specifications. The dependent variable was heart disease prevalence within each census tract. The key independent variable of interest was the minimum distance to a hospital, calculated using the aforementioned GIS modules.

Control variables included various socioeconomic factors, such as median household income, education level, employment rate, and housing characteristics. Additionally, air pollution levels (PM2.5 concentrations) for each census tract were included as control variables, as pollution may play a role in chronic illness. Additionally, control variables relating to health characteristics were included to control comorbidities and other health factors that may influence heart disease prevalence.

Multiple model specifications were used to assess the robustness of the results across our different control variable groups.

Results:

Our first table shows heart disease prevalence in Idaho census tracts with points of hospitals scattered across the state. We see a large number of hospitals concentrated within the southwestern edge of the state in the treasure valley region and continuing east along the magic and sun valleys. We also have hospitals concentrating along the upper panhandle where the population centers of Lewiston and Cour D'Alene, Idaho. Notably, heart disease prevalence seems to be smaller (blue) in census tracts with close proximity to hospital locations. Generally, census tracts that extend out further away from hospital areas seem to have more prevalence of heart disease. It should be noted, however, that some of the census tracts are quite large compared to others, and therefore populations can be very sparse within them (national forest area in the center of state) or be concentrated on a particular side of the area (like a side that is close to a hospital/nearest a population center/or transportation infrastructure). One of the reasons population-based centroids were used rather than geographic centroids was to address these issues with spatial population distribution.



Our next table presents the results of our three model specifications, simple, neighborhood, environment, and health. The first model, Simple, shows output based on one-

	(1) Simple	(2) Neighborhood	(3) Environment	(4) Health
Min Distance to Hospital	0.0822*** (0.0115)	0.00729 (0.00912)	0.00454 (0.00895)	0.000679 (0.00244)
Median Income		-0.0000350*** (0.00000729)	-0.0000342*** (0.00000736)	-0.00000225 (0.00000211)
Median Age		0.110*** (0.0136)	0.111*** (0.0134)	-0.00774 (0.00491)
Pop Female		0.000201 (0.000203)	0.000209 (0.000206)	-0.000125 (0.0000719)
Average Household Size		-0.0805 (0.417)	-0.126 (0.419)	-0.182* (0.0764)
Total Educ Attained		0.00000889 (0.000241)	0.0000163 (0.000246)	0.000194 (0.000109)
Educ Level HS		0.000576 (0.000409)	0.000570 (0.000410)	0.0000370 (0.000125)
Some College		-0.000511 (0.000371)	-0.000498 (0.000375)	-0.000208 (0.000116)
Associate degree		-0.000568 (0.000463)	-0.000550 (0.000470)	-0.000163 (0.000116)
Bachelor's Degree		-0.000418 (0.000285)	-0.000451 (0.000285)	0.0000163 (0.0000692)
Master's Degree		-0.000922 (0.000628)	-0.000936 (0.000637)	-0.000501** (0.000162)
Professional Degree		0.00140 (0.000958)	0.00135 (0.000963)	-0.000181 (0.000289)
Rent Income Ratio		-0.00420** (0.00132)	-0.00404** (0.00130)	-0.000438 (0.000265)
RI Ratio >50%		-0.000809 (0.000606)	-0.000787 (0.000622)	0.0000375 (0.000130)
PM2.5 Concentration			-0.0596 (0.0482)	-0.000216 (0.0172)
Lack of HI access				-0.0521* (0.0236)
Binge Drinking				-0.104*** (0.0186)
Blood Pressure High				0.107*** (0.0232)
Diabetes Prevalence				0.152*** (0.0409)
Obesity Prevalence				-0.103*** (0.0104)
Physical Health not good/fair >14 days				0.292** (0.107)
<i>N</i>	297	297	297	297

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

-independent variable of minimum distance to a hospital. We see that for every mile increase in distance to a hospital, prevalence of heart disease increases by 0.0822. This is a statistically significant result and in accordance with our hypothesis. Economically, it means that adjusting for population size, for every 125 miles away from a census tract a hospital is, we would expect to see one more case of heart disease. This isn't not account for very much in reality as the maximum distance we see within Idaho census minimum distance is around 50 miles away.

Our second model, Neighborhood, variables relating to neighborhood characteristics and socioeconomic factors are introduced. We see that our key independent variable, minimum distance to a hospital, loses statistical significance. Meaning we see no relationship between the two variables of distance and heart disease prevalence. We do see some socio-economic variables have a statistically significant effect on heart disease prevalence such as median income and rent to income ratio.

Our third model, Environment, adds the variable PM2.5 to account for the effect of pollution of the incidence of chronic illnesses like heart disease. We see median age gain significance in this model specification.

Our last model, Health, includes variables relating to health factors that may influence heart disease incidence within Idaho census tracts. Our variable our interest, minimum distance to a hospital, remains statistically insignificant. Several of our previous socio-economic variables lose significance here. We see many of the health factors have significant effects. Counterintuitively, many of them, such as binge drinking and obesity prevalence, show effects contrary to what we may expect in the form of negative effects on our dependent variable. Other factors such as high blood pressure prevalence, diabetes prevalence, and reporting of physical

health not being good or fair for more than 14 days in the year show positive effects on the incidence of heart disease.

Conclusion:

This study examined the relationship between hospital distance and heart disease prevalence in Idaho census tracts, using a combination of geographic information system (GIS) analysis and ordinary least squares (OLS). Our findings show no relationship between hospital distance and heart disease prevalence. In our initial analysis, larger distances to hospitals seemed to be associated with higher rates of heart disease prevalence, supporting our hypothesis. However, with different model specifications controlling for neighborhood, environmental, and health-related variables, the relationship between hospital distance and heart disease prevalence became statistically insignificant. This indicates that other factors, such as health status, or as research suggests, healthcare utilization, among others, may play a more significant role in determining heart disease prevalence than hospital distance.

There were several limitations to the study that should be addressed in future research on the subject. Firstly, distance to a hospital was not measured via currently available transportation networks but rather via a straight line. This could have impacted the results as roads/traffic can impact distance and transportation costs to the hospital. Another limitation was the use of census tract centroids, as they provide the best estimate for average household distance, but cannot properly estimate every household, especially in larger census tracts. Lastly, several control variables such as household car characteristics or other pollution measurement could not be obtained and included in the model specification.

Overall, our study contributes to the understanding of how healthcare access can or cannot influence health outcomes. In the policy sphere, this may provide evidence that the current arrangement of hospitals within Idaho does a sufficient job in providing access to at risk individuals of heart disease. Further research, however, is needed to explore the other mechanisms driving heart disease in our communities.

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